

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week -3

TASK 1 — DOCUMENT MODELLING

Step 1: Open MongoDB Shell and Connect to MongoDB Atlas

```
sahasra@Sah ~ % mongosh "mongodb+srv://cluster0.2tskxjx.mongodb.net/" --apiVersion 1 --username sahasragaya
thrichalla_db_user
Enter password: *****
(Current Mongosh Log ID: 6a83d0fabe050d09dae66695)
Connecting to:   mongodb+srv://<credentials>@cluster0.2tskxjx.mongodb.net/?appName=mongosh+2.9.2
Using MongoDB:  8.0.29 (API Version 1)
Using Mongosh:  2.9.2
mongosh 2.10.0 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/
```

Step 2: Create and Switch to the BookFlow Database

```
Atlas atlas-ij50n7-shard-0 [primary] test> use BookFlow
switched to db BookFlow
```

Step 3: Insert the First Book Document

```
Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.insertOne({
  title: "The Silent Algorithm",
  author: "R. Menon",
  genre: "Technology",
  publishedYear: 2022,
  formats: ["Hardcover", "Digital", "Audiobook"],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: "K. Rao",
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: "M1001",
      rating: 5,
      comment: "Excellent deep dive into digital systems design.",
      date: ISODate("2023-01-15")
    },
    {
      member_id: "M1042",
      rating: 4,
      comment: "Good but the Digital edition had formatting issues.",
      date: ISODate("2023-02-20")
    }
  ]
});
acknowledged: true,
insertedId: ObjectId('6a83d19ebe050d09dae66696')
```

Step 4: Insert the Remaining Book Documents

```
Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.insertMany([
  {
    title: "Ocean of Code",
    author: "A. Fernandes",
    genre: "Technology",
    publishedYear: 2021,
    formats: ["Digital", "Paperback"],
    specs: {
      pages: 210,
      fileSize_MB: 2.1,
      drm: false
    },
    avgRating: 4.2,
    reviews: [
      {
        member_id: "M2001",
        rating: 4,
        comment: "Solid Digital read, clean layout.",
        date: ISODate("2022-05-10")
      },
      {
        member_id: "M2002",
        rating: 5,
        comment: "Best programming book this year.",
        date: ISODate("2022-06-01")
      }
    ]
  },
  {
    title: "Whispers of Yesterday",
    author: "S. Kapoor",
    genre: "Fiction",
    publishedYear: 2019,
    formats: ["Hardcover", "Paperback"],
    specs: {
      pages: 410,
      binding: "Sewn"
    },
    avgRating: 3.8,
    reviews: [
      {
        member_id: "M3001",
        rating: 3,
        comment: "Slow start but strong ending.",
        date: ISODate("2020-01-15")
      },
      {
        member_id: "M3002",
        rating: 4,
        comment: "Beautifully written prose.",
        date: ISODate("2020-02-11")
      }
    ]
  },
  {
    title: "Quantum Horizons",
    author: "R. Menon",
    genre: "Science",
    publishedYear: 2023,
    formats: ["Digital", "Audiobook"],
    specs: {
      fileSize_MB: 5.6,
      drm: true,
      narrator: "P. Iyer",
      runtime_min: 540
    },
    avgRating: 4.9,
    reviews: [
      {
        member_id: "M4001",
        rating: 5,
        comment: "The Digital format made annotations so easy.",
        date: ISODate("2023-08-01")
      },
      {
        member_id: "M4002",
        rating: 5
      }
    ]
  }
]);
```

```

|         rating: 5,
|         comment: "Mind-bending and well narrated.",
|         date: ISODate("2023-09-14")
|       }
|     ]
|   }
| })
|
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a83d206be050d09dae66697'),
    '1': ObjectId('6a83d206be050d09dae66698'),
    '2': ObjectId('6a83d206be050d09dae66699')
  }
}
}

```

Step 5: Verify the Inserted Book Documents

```

Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.find()
[
  {
    _id: ObjectId('6a83d19ebe050d09dae66696'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a83d206be050d09dae66697'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      }
    ]
  }
]

```

TASK 2 — ADVANCED QUERYING

Step 1 — Find books with an average rating above 4

```

Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.find(
|   { avgRating: { $gt: 4 } },
|   { title: 1, avgRating: 1, _id: 0 }
| )
[
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'Ocean of Code', avgRating: 4.2 },
  { title: 'Quantum Horizons', avgRating: 4.9 }
]

```

Step 2 — Find books available in Digital or Audiobook formats

```

Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.find(
|   { formats: { $in: ["Digital", "Audiobook"] } },
|   { title: 1, formats: 1, _id: 0 }
| )
[
  {
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  { title: 'Ocean of Code', formats: [ 'Digital', 'Paperback' ] },
  { title: 'Quantum Horizons', formats: [ 'Digital', 'Audiobook' ] }
]
Atlas atlas-ij50n7-shard-0 [primary] BookFlow> █

```

Step 3: Finding books with Digital format using a case-insensitive regular expression

```
Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.find(
|   { formats: { $regex: "Digital", $options: "i" } },
|   { title: 1, formats: 1 }
| )
[ ]
[
  {
    _id: ObjectId('6a83d19ebe050d09dae66696'),
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  {
    _id: ObjectId('6a83d206be050d09dae66697'),
    title: 'Ocean of Code',
    formats: [ 'Digital', 'Paperback' ]
  },
  {
    _id: ObjectId('6a83d206be050d09dae66699'),
    title: 'Quantum Horizons',
    formats: [ 'Digital', 'Audiobook' ]
  }
]
```

Step 4 — Search reviews containing "Digital"

```
Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.find(
|   { "reviews.comment": { $regex: "Digital", $options: "i" } },
|   { title: 1, "reviews.$": 1 }
| )
[ ]
[
  {
    _id: ObjectId('6a83d19ebe050d09dae66696'),
    title: 'The Silent Algorithm',
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a83d206be050d09dae66697'),
    title: 'Ocean of Code',
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a83d206be050d09dae66699'),
    title: 'Quantum Horizons',
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      }
    ]
  }
]
```

TASK 3 — AGGREGATION PIPELINE: TOP RATED REPORT

STEP 1 — \$match: Filter books published after 2020

```
Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.aggregate([
| {
|   $match: {
|     publishedYear: { $gt: 2020 }
|   }
| },
| ])
[
| {
|   _id: ObjectId('6a83d19ebe050d09dae66696'),
|   title: 'The Silent Algorithm',
|   author: 'R. Menon',
|   genre: 'Technology',
|   publishedYear: 2022,
|   formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
|   specs: {
|     pages: 340,
|     fileSize_MB: 4.2,
|     drm: true,
|     narrator: 'K. Rao',
|     runtime_min: 610
|   },
|   avgRating: 4.6,
|   reviews: [
|     {
|       member_id: 'M1001',
```

Step 2: Separating Embedded Reviews Using \$unwind

```
Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.aggregate([
| {
|   $match: {
|     publishedYear: { $gt: 2020 }
|   }
| },
| {
|   $unwind: "$reviews"
| }
| ])
[
| {
|   _id: ObjectId('6a83d19ebe050d09dae66696'),
|   title: 'The Silent Algorithm',
|   author: 'R. Menon',
|   genre: 'Technology',
|   publishedYear: 2022,
|   formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
|   specs: {
|     pages: 340,
|     fileSize_MB: 4.2,
|     drm: true,
|     narrator: 'K. Rao',
|     runtime_min: 610
|   },
|   avgRating: 4.6,
```

Step 3: Calculating the Average Review Rating for Each Book Using \$group

```
Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.aggregate([
| {
|   $match: {
|     publishedYear: { $gt: 2020 }
|   }
| },
| {
|   $unwind: "$reviews"
| },
| {
|   $group: {
|     _id: "$title",
|     averageReviewRating: {
|       $avg: "$reviews.rating"
|     }
|   }
| }
| ])
[
| {
|   _id: 'The Silent Algorithm', averageReviewRating: 4.5 },
| {
|   _id: 'Quantum Horizons', averageReviewRating: 5 },
| {
|   _id: 'Ocean of Code', averageReviewRating: 4.5 }
| ]
```

Step 4: Sorting Books by Average Review Rating Using \$sort

```
Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.aggregate([
| {
|   $match: {
|     publishedYear: { $gt: 2020 }
|   }
| },
| {
|   $unwind: "$reviews"
| },
| {
|   $group: {
|     _id: "$title",
|     averageReviewRating: {
|       $avg: "$reviews.rating"
|     }
|   }
| },
| {
|   $sort: {
|     averageReviewRating: -1
|   }
| }
| ])
[
| {
|   _id: 'Quantum Horizons', averageReviewRating: 5 },
| {
|   _id: 'The Silent Algorithm', averageReviewRating: 4.5 },
| {
|   _id: 'Ocean of Code', averageReviewRating: 4.5 }
| ]
```

Step 5: Generating the Final Top Rated Books Report Using \$project

```
-Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.aggregate([
| {
|   $match: {
|     publishedYear: { $gt: 2020 }
|   },
|   {
|     $unwind: "$reviews"
|   },
|   {
|     $group: {
|       _id: "$title",
|       averageReviewRating: {
|         $avg: "$reviews.rating"
|       }
|     }
|   },
|   {
|     $sort: {
|       averageReviewRating: -1
|     }
|   },
|   {
|     $project: {
|       _id: 0,
|       title: "$_id",
|       averageReviewRating: 1
|     }
|   }
| ]])
[
| { averageReviewRating: 5, title: 'Quantum Horizons' },
| { averageReviewRating: 4.5, title: 'The Silent Algorithm' },
| { averageReviewRating: 4.5, title: 'Ocean of Code' }
| ]
```

TASK 4 — INDEXING FOR SPEED

Step 1: Creating a Text Index on Review Comments

```
[Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.createIndex({ "review|
| s.comment": "text" })
| reviews.comment_text
```

Step 2: Searching Review Comments Using the Text Index

```
Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.find(
| { $text: { $search: "Digital" } },
| { title: 1, score: { $meta: "textScore" } }
| ).sort({ score: { $meta: "textScore" } })
[
| {
|   _id: ObjectId('6a83d19ebe050d09dae66696'),
|   title: 'The Silent Algorithm',
|   score: 1.1833333333333333
| },
| {
|   _id: ObjectId('6a83d206be050d09dae66699'),
|   title: 'Quantum Horizons',
|   score: 0.6
| },
| {
|   _id: ObjectId('6a83d206be050d09dae66697'),
|   title: 'Ocean of Code',
|   score: 0.6
| }
| ]
```

Step 3: Verifying Text Index Usage Using .explain()

```
Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.find(
| { $text: { $search: "Digital" } }
| ).explain("executionStats")
{
|   explainVersion: '1',
|   queryPlanner: {
|     namespace: 'BookFlow.book_metadata',
|     parsedQuery: {
|       '$text': {
|         '$search': 'Digital',
|         '$language': 'english',
|         '$caseSensitive': false,
|         '$diacriticSensitive': false
|       }
|     },
|     indexFilterSet: false,
|     queryHash: 'CF6F4CEE',
|     planCacheShapeHash: 'CF6F4CEE',
|     planCacheKey: '08852285',
|     optimizationTimeMillis: 0,
|     maxIndexedOrSolutionsReached: false,
|     maxIndexedAndSolutionsReached: false,
|     maxScansToExplodeReached: false,
|     prunedSimilarIndexes: false,
|     winningPlan: {
|       isCached: false,
|       stage: 'TEXT_MATCH',
|       indexPrefix: {},
|       indexName: 'reviews.comment_text',
|       parsedTextQuery: {
|         terms: [ 'digit' ],
|         negatedTerms: [],
|         phrases: [],
|         negatedPhrases: []
|       },
|       textIndexVersion: 3,
|       inputStage: {
|         $equi
```

Step 4: Comparing Text Search with a Regular Expression Scan

```
Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.find(
| { "reviews.comment": { $regex: "Digital", $options: "i" } }
| ).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
    indexFilterSet: false,
    queryHash: '9DF7CB16',
    planCacheShapeHash: '9DF7CB16',
    planCacheKey: 'A9183C86',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
      direction: 'forward'
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 0,
    totalKeysExamined: 0,
    totalDocsExamined: 4,
    executionStages: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 5,
      advanced: 3,
      needTime: 1,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,

```

Step 5: Creating a Multikey Index on the Formats Array

```
Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.createIndex({formats:
1 })
formats 1
```

Step 6: Creating an Index on Average Rating

```
Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.createIndex({ avgRati
ng: -1 })
avgRating_-1
```

Step 7: Verifying Index Usage for Rating-Based Queries

```
Atlas atlas-ij50n7-shard-0 [primary] BookFlow> db.book_metadata.find(
| { avgRating: { $gt: 4 } }
| ).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: { avgRating: { '$gt': 4 } },
    indexFilterSet: false,
    queryHash: '85BA8276',
    planCacheShapeHash: '85BA8276',
    planCacheKey: '6EB4A3FD',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      inputStage: {
        stage: 'IXSCAN',
        keyPattern: { avgRating: -1 },
        indexName: 'avgRating_-1',
        isMultiKey: false,
        multiKeyPaths: { avgRating: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: { avgRating: [ '[inf.0, 4)' ] }
      },
      rejectedPlans: []
    },
    rejectedPlans: []
  },

```